



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

॥ वसुधैव कुटुम्बकम् ॥ Re-Accredited by NAAC with 'A++' Grade



Computer Networks/0705210503

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Internet Protocol (IP)

The Internet Protocol is responsible for addressing and routing packets across networks.

Layer: It operates at the **Network Layer (Layer 3)** of the OSI model.

Functions:

1. Addressing hosts uniquely.
2. Packet forwarding and routing between networks.
3. Fragmentation and reassembly of packets.

Versions:

IPv4: Older, widely used.

IPv6: Newer, designed to solve IPv4 limitations.



IPv4 Addressing

- It is one of the core protocols in the Internet layer of the TCP/IP suite.
- It defines how data packets are addressed and routed so that they can travel across networks and reach the correct destination.
- IPv4 is a **32-bit** address scheme, meaning it can provide about **4.3 billion** unique addresses.
- Each address identifies a unique device or host on a network.
- Usually written in dotted-decimal format for human readability. It identifies devices (hosts) on a network.

Example:

Binary:

11000000.10101000.00000001.00000001

Dotted-decimal format: An IPv4 address consists of 32 bits, divided into 4 octets (**8 bits each**).

192.168.1.1



Structure of IPv4 Address

An IPv4 address consists of **4 octets (8 bits each)**, separated by dots (.). Each octet ranges from **0 to 255**.

Example:

172.16.254.1

Breaking it into binary:

172 → 1010110016

16 → 00010000254

254 → 111111101

1 → 00000001



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IPv4 addresses are divided into two parts:

Network part – Identifies the specific network.

Host part – Identifies a device within that network.

Example:

For the address **192.168.1.10** with subnet mask **255.255.255.0**, the breakdown is:

Network portion: 192.168.1

Host portion: 10

The subnet mask tells how many bits are for the network and how many for the host.



Subnet Mask

A subnet mask defines which portion of an IP address refers to the network and which refers to the host.

Example:

IP address: 192.168.1.100

Subnet mask: 255.255.255.0

Convert subnet mask to binary:

255.255.255.0 -> 11111111.11111111.11111111.00000000

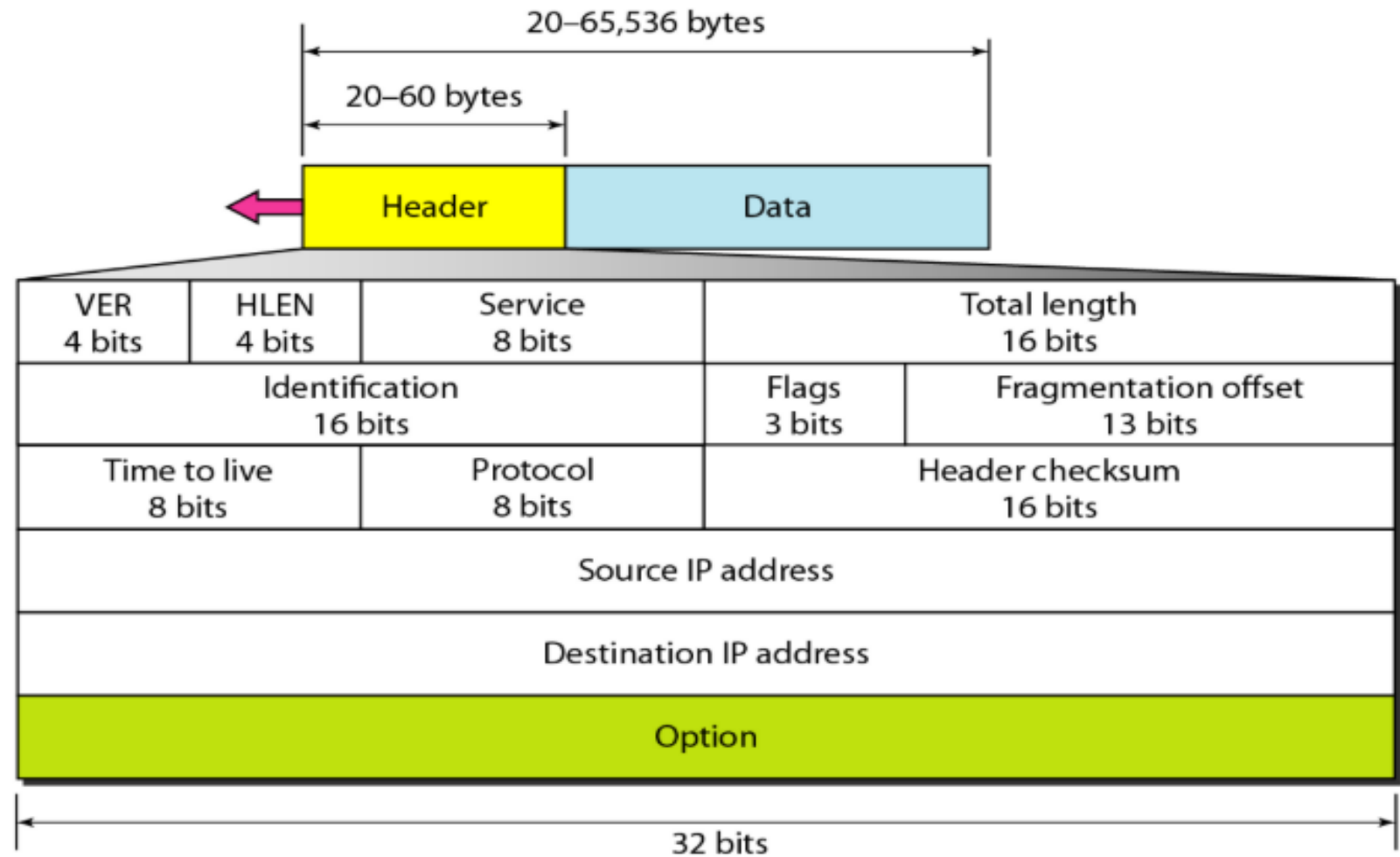
The first **24 bits** are the network, last **8 bits** are for host identification.



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IPV4 Datagram Format





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Field	Size (bits)	Description
Version	4	IP version (4 for IPv4)
HLEN	4	Length of the header in 32-bit words (minimum 5, maximum 15)
Type of Service (ToS)	8	Specifies priority and handling (e.g., delay, throughput)
Total Length	16	Total length of datagram (header + data) in bytes
Identification	16	Unique ID for reassembling fragmented datagrams
Flags	3	Controls fragmentation; e.g., "Don't Fragment" flag
Fragment Offset	13	Position of fragment in the original datagram
Time to Live (TTL)	8	Limits lifespan to prevent routing loops
Protocol	8	Specifies upper-layer protocol (TCP, UDP, ICMP, etc.)
Header Checksum	16	Error-checking of header data
Source Address	32	IPv4 address of the sender
Destination Address	32	IPv4 address of the receiver
Options	Variable	Optional settings (rarely used)



Classes of IPv4 Addresses

IPv4 addresses are traditionally divided into classes based on the first few bits to define the network size and structure.

Class A

Range: 0.0.0.0 to 127.255.255.255

First octet range: 1–126

Network bits: 8

Host bits: 24

Default Subnet Mask: 255.0.0.0

Usage: Large networks (e.g., big organizations)

Example:

10.0.0.1

Network portion → 10

Host portion → 0.0.1



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Class B

Range: 128.0.0.0 to 191.255.255.255

First octet range: 128–191

Network bits: 16

Host bits: 16

Default Subnet Mask: 255.255.0.0

Usage: Medium-sized networks (e.g., universities)

Example:

172.16.32.5

Network portion → 172.16

Host portion → 32.5



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Class C

Range: 192.0.0.0 to 223.255.255.255

First octet range: 192–223

Network bits: 24

Default Subnet Mask: 255.255.255.0

Host bits: 8

Usage: Small networks (e.g., home networks)

Example:

192.168.1.15

Network portion → 192.168.1

Host portion → 15



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Class D

Used for multicast groups

Range: 224.0.0.0 to 239.255.255.255

Class E

Range: 240.0.0.0 to 255.255.255.255

Reserved for future use or experimental purposes

CIDR Notation

CIDR (Classless Inter-Domain Routing) is a modern way of representing the network prefix.

Example: 192.168.1.10/24

This means:

- IP address: 192.168.1.10
- The first 24 bits represent the network portion.

This notation is more flexible than old class-based methods.

IPv4 addresses are divided into classes based on their leading **bits** and **range**.

Class	First Octet Range	Network Bits	Host Bits	Default Subnet Mask	Use Case
A	0–127	8	24	255.0.0.0	Large networks
B	128–191	16	16	255.255.0.0	Medium networks
C	192–223	24	8	255.255.255.0	Small networks
D	224–239	Multicast	-	-	Multicast group
E	240–255 8	Experimental	-	-	Research only

Given 192.168.1.0/24, how many usable host addresses are available?

Total host bits = 8

Total addresses = $2^8 = 256$

Usable addresses = $256 - 2 = 254$ (subtract network and broadcast addresses)

So, you can assign IPs from 192.168.1.1 to 192.168.1.254.